

I-BEAMS

DESIGN OF BEAMS

Design formulae for beams

	Notation
L	= length of span in millimetres
W	= total distributed or point load in Newtons
W_1 or W_2	= point load in Newtons
Σ	= resultant of point loads in Newtons
R_A, R_B, R_C etc.	= reaction at A, B, or C etc, in Newtons
F	= shearing force in Newtons
m	= applied moment in Newton millimetres
M_x	= bending moment in Newton millimetres at distance X from the left hand support A
δ_x	= deflection in millimetres
i_x	= slope in radians
M_A or M_B	= end fixing moment in Newton millimetres
M_{max}	= maximum bending moment in Newton millimetres
$M_{max\ max}$	= absolute maximum bending moment in Newton millimetres
M_{load}	= bending moment under the load in Newton millimetres
δ_{max}	= maximum deflection in millimetres
$\delta_{max\ max}$	= absolute maximum deflection in millimetres
$\delta_{negative}$	= negative (i.e. upward) deflection in millimetres
i_A or i_B	= slope at A or at B in radians
E	= modulus of elasticity, $2.1 \times 10^5 \text{ N/mm}^2$
I	= constant moment of inertia of uniform section beam in mm^4
Sign convention	
Loads	+ positive when acting downward
Support Reaction	+ positive when acting upward
Shearing Force	+ positive on a section where the upward left hand support reaction is greater than the algebraic sum of external loads located left of that section
Bending Moment	+ positive (shown above base line on diagrams) when causing convexity downward
Deflection	+ positive when downward
Slope	appropriate values in radians are given, but the signs depend upon which support or which section is being considered, and can be readily ascertained by inspection

Where space permits, general equations for M_x and i_x at any point of the beam, and also the equation to the elastic line (δ_x), have been included.

Values for Slope. These may be used in evaluating the angle of rotation for rubber bearings and similar constructional elements.